

SHA256 Hash Generator Demo: A Lightweight Web App for Cryptography Learning

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GitHub Repository: <https://github.com/nixonmwangi/HashGenerator.git>

Abstract

We present SHA256 Hash Generator Demo, a simple, browser-based application for generating SHA256 hashes of arbitrary text input. The app is designed for educational and research purposes, providing a lightweight and interactive tool for understanding cryptographic hashing. It requires no backend server, runs entirely in the browser, and is open-source under the MIT License. The source code is available on GitHub and archived with DOI on Zenodo for citation.

1. Introduction

Cryptographic hash functions are fundamental in modern cybersecurity, used in password storage, digital signatures, and blockchain technologies. To aid learning and experimentation, we introduce a minimalistic web application that allows users to input text and instantly observe the corresponding SHA256 hash. This tool is ideal for students, researchers, or developers exploring cryptographic concepts.

2. Implementation

- Language/Framework: HTML, CSS, JavaScript
- Crypto API: Web Crypto API (built into all modern browsers)
- Usage: Users type text into the input area, click "Generate Hash", and receive the SHA256 hash in hexadecimal format.
- Cross-platform: Works on desktop, tablet, and mobile devices without any installation.

Example code snippet:

```
const encoder = new TextEncoder();  
const data = encoder.encode(inputText);  
const hashBuffer = await crypto.subtle.digest('SHA-256', data);
```

3. Availability

- Source code: <https://github.com/nixonmwangi/HashGenerator.git>
- Live demo: GitHub Pages link
- License: MIT License
- Archival DOI (optional): <https://doi.org/10.5281/zenodo.xxxxxx>

4. Conclusion

This lightweight SHA256 Hash Generator Demo provides a simple, interactive environment to explore cryptographic hashing. Its minimalistic design ensures accessibility for educational purposes while maintaining open-source availability for research and experimentation.